

Jeffrey Li

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Education

Worcester Polytechnic Institute (WPI)

Worcester, MA

B.S. and M.S. in Computer Science

Expected May 2026

Relevant Coursework: Software Engineering, Object-Oriented Analysis and Design, Data Structures and Algorithms, Machine Learning, Database Systems, Natural Language Processing, Operating Systems, Computer Networks

Work Experience

Ingredion Incorporated, *Software Engineer Intern*

May 2025 – August 2025

- Designed and implemented a LabVIEW application for real-time defect detection in starch filtration machinery.
- Built and deployed a real-time audio classification pipeline using PyTorch and a 1D CNN to detect mechanical anomalies.
- Developed a signal processing pipeline using SciPy and Librosa (STFT, Mel spectrograms, MFCCs) for feature extraction from audio data.
- Implemented data preprocessing and feature engineering using NumPy and Pandas.
- Trained and validated models achieving 98% accuracy, contributing to improved defect detection and reduced manual inspection effort.

Acton Institute of Computer Science, *Full-Stack Developer*

June 2023 – August 2023

- Developed a full-stack web application for summer camp registration using React, Express, and Firebase.
- Implemented user authentication, registration workflows, and data persistence for 130+ user accounts.
- Deployed the application to Netlify, supporting 80+ successful camp registrations.
- Collaborated in a team of 5 in an Agile (Scrum) environment, delivering features through iterative development cycles.

Projects

Personal Project - Running Log Application

- Built a full-stack running log application using Next.js, PostgreSQL, and Prisma, featuring authenticated CRUD workflows and role-based route protection.
- Designed a normalized relational database schema with indexed queries supporting weekly mileage aggregation, calendar heatmaps, and shoe-level distance analytics.
- Implemented data visualizations and exports (weekly trends, consistency metrics, CSV export) with a responsive UI using React and Tailwind CSS.

Software Engineering Course Project - Hospital Application in Collaboration with Brigham and Women's Hospital

- Collaborated in a team of 11 to develop a cloud-based hospital navigation and service request system using React, TypeScript, and PostgreSQL.
- Contributed to frontend architecture and database design supporting real-time pathfinding, voice navigation, and service request automation.
- Implemented authentication using Auth0 and deployed infrastructure using AWS EC2 and RDS.

Skills

- **Languages:** Python, Java, TypeScript/JavaScript, SQL, C/C++
- **Frameworks & Tools:** React, Next.js, Node.js, Express, Docker, Git/GitHub
- **Databases:** PostgreSQL, MongoDB
- **Machine Learning:** PyTorch, NumPy, Pandas, SciPy, Librosa, Scikit-learn
- **Auth & APIs:** RESTful APIs, NextAuth, Auth0
- **Development Practices:** Agile (Scrum)
- **Other Technical Skills:** LabVIEW

Teaching/Leadership Experience

Worcester Polytechnic Institute, *Peer Learning Assistant*

August 2023 – Present

- Supported instruction for various math courses, leading weekly recitation sections for 25-30 students.
- Conducted office hours, tutored students and graded coursework for 100+ students.
- Collaborated with faculty and peers to enhance course delivery and improve student outcomes